Secure API Gateway Management System

Project Overview

The Secure API Gateway Management System provides a centralized gateway through which every API request passes before reaching backend services. The gateway authenticates requests using JWT, verifies access permissions, applies rate limits, records request and response logs, and routes valid requests to the appropriate service. This improves security, monitoring, and centralized API management.

Problem Statement

Modern applications expose multiple APIs that can be accessed by different users and applications. Without a centralized gateway, maintaining security, monitoring API traffic, and controlling access becomes difficult. This project addresses these challenges by introducing a secure API Gateway.

Proposed Solution

The system introduces a single gateway responsible for validating requests, enforcing security policies, maintaining logs, and forwarding only valid requests. Administrators can manage APIs, monitor traffic, and configure gateway rules through a web dashboard.

Project Workflow

User Login → JWT Generation → API Request → JWT Validation → Authorization Check → Rate Limit Check → Request Logging → Route to Backend Service → Receive Response → Response Logging → Return Response to User.

Core Modules

• User Authentication
• API Management
• Request Routing
• Rate Limiting
• Request & Response Logging
• Dashboard & Analytics
• User Management
• Admin Panel

Technology Stack	Database Tables
React.js	Users
Spring Boot	API_Keys
Gradle	APIs
PostgreSQL	Request_Logs
JWT	Rate_Limits
Postman, Git, GitHub	

Expected Outcome

The completed system demonstrates secure API access, centralized request management, monitoring, and routing using an enterprise-inspired architecture. It highlights backend security concepts and API lifecycle management through a clean and scalable implementation.